

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Winter 2017 – 18 Examination

Semester: 3
Subject Code: 03105202
Subject Name: Operating System

Date: 28/12/2017
Time: 10:30 am to 1:00 pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions: (All are compulsory) (Each of one mark) **(15)**

1. Interval between the time of submission and completion of the job is called _____. (Waiting time, Turnaround time, Throughput, Response time)
2. When CPU switches from one process to another process, the system must _____ the context of the old process in its PCB, and loads the saved context for the new process.
3. Semaphore is a _____ tool.
4. Each process in a system has a segment of code, called _____, in which the process may be changing common variables, updating a table, writing a file. (Critical section, semaphore, race condition)
5. MMU is a hardware device that provides runtime mapping from _____ to _____.
6. Out of four conditions, the three conditions are given below.
(a) Circular Wait
(b) Hold and Wait
(c) Mutual exclusion
Which conditions is **REMAINING** for necessary and sufficient conditions for deadlock?
Ans: _____
7. The TestAndSet instruction is executed :
a) after a particular process
b) periodically
c) atomically
d) None of these
8. Pages are only loaded when they are demanded during program execution is called _____.
9. If the no. of pages=2K, Page size=2K and Physical address is of 18 bits then no. of frames are _____.
10. If the system having a Page table with 4K entries, LA is of 29 bits and No. of frames are 512 then PA is of _____ bits.
11. **Answer True/ False:** Device driver which is the standard interface through which the device controller communicates with the Operating Systems.
12. **Answer True/ False:** In acyclic graph same file or sub-directories cannot be linked with two different directories
13. **Answer True/ False:** Computer worms spread much more rapidly than computer viruses.

14. In Access Control Matrix Row is represented by _____ and Column is represented by _____.

15. Trojan horse are _____ programs placed inside a legitimate application (authorized, unauthorized).

Q.2 Answer the following questions. (Attempt any three)

(15)

A) Explain working of DMA with Diagram.

B) Which are the three requirements that must be satisfied to solve critical-section Problem? Explain each of them.

C) Differentiate between Compiler and Interpreter and also write difference between Program and Process.

D) Describe advantages and disadvantages of Time Sharing Systems (TSS).

Q.3 A) (i) Draw the Process State Diagram.

(07)

(ii) Solve Example using Round Robin (RR) Scheduling Algorithm. Time Quantum = 4. Find Average Waiting time and Average Turnaround time.

Process	Arrival Time (ms)	Burst Time (ms)
P1	0	24
P2	1	3
P3	2	3

(08)

B) Consider following reference string 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3. Number of page frames are 5 and all are empty initially. Calculate no of page faults for FIFO, LRU and OPTIMAL page replacement algorithms.

OR

B) Suppose that a disk drive has 200 cylinders, numbered 0 to 199. The Head Pointer is currently at 53. The queue of pending requests, in FIFO order, is 98, 183, 37, 122, 14, 124, 65, 67. Starting from the current head position, what is the **Average seek distance** (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk scheduling algorithm. 1) FCFS 2) SCAN

(08)

Q.4 A) (i) Differentiate External fragmentation and internal fragmentation.

(07)

(ii) Differentiate Paging and Segmentation.

OR

A) Explain any two File allocation Methods with appropriate diagram.

(07)

B) (i) Explain Producer Consumer Problem.

(08)

(ii) Consider a system with Five Processes P0 through P4 and three resources A, B and C. Resource type A, B and C have total resources 10, 5 and 7 instances respectively. Suppose at time T0 the following snapshot of the system has been taken. Find need matrix. Also find whether the system is in safe state or not? Find out the safe sequence.

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			